it simply happened. This bugged Newton—not because it was illogical or incompatible with observation, but just because it seemed wrong. Spooky, one might say.

> It is inconceivable that inanimate brute matter should, without the Mediation of something else which is not material, operate upon and affect other matter without mutual contact. . . . Gravity must be caused by an agent acting constantly according to certain laws; but whether this agent be material or immaterial, I have left to the consideration of my readers.

There is indeed an "agent" that causes gravity to act the way it does, and that agent is perfectly material—it's the gravitational field. This concept was first introduced by Pierre-Simon Laplace, who was able to rewrite Newton's theory of gravity so that the force was carried by a gravitational potential field, rather than simply hopping mysteriously across infinite distances. But a change in the force still happened instantaneously through all of space. It wasn't until Einstein came along with general relativity that changes in the gravitational field, just like changes in the electromagnetic field, were shown to travel through space at the speed of light. General relativity replaces Laplace's potential with the "metric" field, a mathematically sophisticated way of characterizing the curvature of spacetime, but the general idea of a gravitational field pervading all of space has remained intact.

The idea of a field carrying a force is conceptually appealing because it instantiates the idea of locality. As the Earth moves, the direction of its gravitational pull doesn't change instantly throughout the universe. Rather, it changes right where the Earth is located, and then the field at that point tugs on the field nearby, which tugs on the field a little farther away, and so on in a wave moving outward at the speed of light.

Modern physics extends this idea to literally everything in the universe. The Core Theory is constructed by starting with a set of fields and then quantizing them. Even particles like electrons and quarks are really vibrations in quantum fields. That's a wonderful story all by itself, but our aim in this chapter is slightly more modest: to understand the "vacuum" in quantum field theory, the quantum state corresponding to empty space. (I've relegated a brief discussion of states with actual particles in them to the Appendix.) Later we'll tackle the quantum emergence of space itself, but for now we'll be drearily conventional and think about quantum field